培道中學校

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科目：概率論與隨機過程 2019010996

$$P(S=s\mid A=a)=\begin{cases}\dfrac{\binom{m}{s}p^{2s}\cdot\binom{m-s}{a-2s}\left[2p(1-p)\right]^{a-2s}\cdot(1-p)^{2[m+s-a]}}{\binom{2m}{a}p^{a}(1-p)^{2m-a}}, & 0\le 2s\le a\\[2ex] 0, \text{ otherwise}\end{cases}$$

$$=\begin{cases}\dfrac{m!}{s!\cancel{(m-s)!}}\cdot\dfrac{\cancel{(m-s)!}}{(a-2s)!(m+s-a)!}\cdot 2^{a-2s}\cdot\dfrac{a!(2m-a)!}{(2m)!}, & 0\le 2s\le a\\[2ex] 0, \text{ otherwise.}\end{cases}$$

$$sP(S=s\mid A=a)=\begin{cases}\dfrac{m\cdot(m-1)!\cdot 2^{(a-2)-2(s-1)}}{(s-1)!\left[(a-2)-2(s-1)\right]!\left[(m-1)+(s-1)-(a-2)\right]!}\cdot\dfrac{a(a-1)(a-2)!\left[2(m-1)-(a-2)\right]!}{2m\cdot(2m-1)(2m-2)!}, & 2\le 2s\le a\\[2ex] 0, \text{ otherwise}\end{cases}$$

$$=\begin{cases}\dfrac{\cancel{m}\cdot a(a-1)}{2\cancel{m}\cdot(2m-1)}P(S=s-1\mid A=a-2), & 2\le 2s\le a\\[1ex] 0, \text{ otherwise}\end{cases}$$

$$E[S\mid A=a]=\sum_{2\le 2s\le a}sP(S=a\mid A=a)=\frac{a(a-1)}{2(2m-1)}$$

(a) Denote the area of subset S is S_0

$$E[X_i]=1\times\frac{S_0}{1}+0\times\frac{1-S_0}{1}=S_0$$

$$\Rightarrow E[S_n]=E\left[\frac{X_1+X_2+\cdots+X_n}{n}\right]=\frac{1}{n}\sum_{i=1}^{n}E[X_i]=S_0$$

(b) $\operatorname{var}(S_n)=E[S_n^2]-\left(E[S_n]\right)^2$

$$=\frac{1}{n^2}\left[\sum_{i=1}^{n}E[X_i^2]+2\sum_{1\le i<j\le n}E[X_iX_j]\right]-S_0^2$$

$$=\frac{1}{n^2}\left[nS_0+n(n-1)S_0^2\right]^2-S_0^2=\frac{S_0}{n}+\left(1-\frac{1}{n}\right)S_0^2-S_0^2=\frac{S_0(1-S_0)}{n^2}\xrightarrow[(n\to\infty)]{}0$$

(b) $$nS_n = \sum_{k=1}^{n} X_k = \sum_{k=1}^{n-1} X_k + X_n = (n-1)S_{n-1} + X_n$$

$$\Rightarrow S_n = \left(1-\frac{1}{n}\right) S_{n-1} + \frac{1}{n} X_n$$

(c)

```
main.cpp
              Code, Compile, Run and Debug C++ program online.
Write your code in this editor and press "Run" button to compile and execute it.

*******************************************************************************/

#include <iostream>
#include <cstdlib>
#include <cmath>
using namespace std;

int main()
{
    unsigned int X[10001];
    float P[10001][2], S[10001],max=0;
    srand(0);
    S[0]=0;
    for(int k=1; k<=10000;k++){
        P[k][0]=rand()/double(RAND_MAX)-0.5;
        P[k][1]=rand()/double(RAND_MAX)-0.5;
        X[k]=(P[k][0]*P[k][0]+P[k][1]*P[k][1] <= 0.25) ? 1 : 0;
        S[k]=S[k-1]+(X[k]-S[k-1])/k;
        cout << "S[" << k <<"]=" << S[k] << endl;
    }
    return 0;
}
```

```
S[9997]=0.779836
S[9998]=0.779858
S[9999]=0.77988
S[10000]=0.779902


...Program finished with exit code 0
Press ENTER to exit console.
```

Let the result above multiply 4, we shall get the value of pi.

(d)

```
main.cpp
#include <cmath>
using namespace std;
#define PI 3.1415926
int main()
{

    srand(0);
    unsigned int X[10001];
    float P[10001][2],S[10001],x,y,max=0;
    S[0]=0;
    for(int n=1; n <=10000; n++){
        x=rand();y=rand();
        if(max < abs(x) || max < abs(y))
            max = (abs(x) > abs(y)) ? abs(x) : abs(y);
        P[n][0]=x;P[n][1]=y;
    }
    for(int n=1; n <=10000; n++){
        P[n][0] /= max;
        P[n][1] /= max;
        X[n] = (cos(PI*P[n][0])+sin(PI*P[n][1])<=1&&cos(PI*P[n][0])+sin(PI*P[n][1])>=0) ? 1 : 0;
        S[n] = S[n-1] + (X[n]-S[n-1])/n;
        cout << "S[" << n << "]=" << S[n] << endl;
    }
    return 0;
}
```

```
S[9997]=0.373912
S[9998]=0.373875
S[9999]=0.373937
S[10000]=0.374


...Program finished with exit code 0
Press ENTER to exit console.
```

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3.(a) Denote the number of days which elapse between the acquisitions of the jth new type of object and the (j+1)th new type is $D_{(j,j+1)}$

$$E[D_{(j,j+1)}] = \sum_{n=1}^{\infty} n\left(\frac{j}{c}\right)^{n-1}\left(1-\frac{j}{c}\right) = \left(1-\frac{j}{c}\right)\sum_{n=1}^{\infty} n\left(\frac{j}{c}\right)^{n-1}$$

$$= \left(1-\frac{j}{c}\right)\left[\frac{1}{1-\frac{j}{c}} + \frac{j}{c}\,\frac{E[D_{(j,j+1)}]}{1-\frac{j}{c}}\right] = 1+\frac{j}{c}E[D_{(j,j+1)}]$$

$$E[D_{(j,j+1)}] = \frac{1}{1-\frac{j}{c}} = \frac{c}{c-j}$$

(b) Denote the number of days which elapse before I have a full set of objects. is N

$$E[N] = \sum_{j=0}^{c-1} E[D_{(j,j+1)}] = \sum_{j=0}^{c-1}\frac{c}{c-j}$$

4.(a) Denote $\vec{N} = (N_1, N_2, \ldots, N_t)$, $\vec{n} = (n_1, n_2, \ldots, n_t)$, $\sum_{k=1}^{t} n_k = n$

$\cancel{P(\vec{N}=\vec{n}) = \frac{n!}{\prod_{k=1}^{t} n_k!} p_1^{n_1} p_2^{n_2} \cdots p_t^{n_t}}$ $P(\vec{N}=\vec{n}) = n!\prod_{k=1}^{t}\left(\frac{p_k^{n_k}}{n_k!}\right) = \frac{n!}{n_1!n_2!\cdots n_t!}p_1^{n_1}p_2^{n_2}\cdots p_t^{n_t}$

(b) $\cancel{P_{N_i}(n_i) \binom{n}{n_i} p_i^{n_i}(1-p_i)^{n-n_i}}$

$$\text{(b)} = P_{N_i}(n_i) = \begin{cases} \binom{n}{n_i} p_i^{n_i}(1-p_i)^{n-n_i}, & 0 \le n_i \le n \\ 0, & \text{otherwise} \end{cases}$$